

A Guide To MODEL ROCKETRY



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Model rocketry is an interesting hobby. To have a safe and predictable flight, rocketeers must have keen interest and some experience in the field. Lack of insight into the underlying technology reflects badly on them.

Model rocket launch system

Every model rocket flight begins with a single event; an igniter is electrically initiated, which, in turn, ignites the black powder or composite propellant. To have a guaranteed flight, the rocket must be held in position before launch and guided during the first fraction of a second of flight until it is going fast enough for its fins to keep it flying in the proper direction.

The second function is to provide adequate electrical current to cause engine ignition. So the launch system for a model rocket is designed to do two important jobs. The first job, handled by the launchpad, is to hold the rocket before and during launch, and the second job, cared by the launch controller, is to ignite the rocket engine/motor.

The launchpad is usually designed to aim the rocket straight up, and is adjustable so that the model rocket can be aimed either vertically or within 30° to any side of vertical to correct for wind conditions. A small tube, called launch lug, on the model rocket fits over the launch rod to keep the rocket straight on the rod. By the time the model rocket's launch lug leaves the launch rod, the rocket is already going fast enough

for its fins to provide adequate guidance to keep it moving in the desired direction.

The igniter, held firmly in place with an igniter plug, is placed with its bend all the way to the bottom of the nozzle and firmly in contact with the propellant. Electrical current delivered by the launch controller is adequate to heat the igniter, which causes the propellant to ignite. The thermoplastic coating on the end of the igniter helps protect against igniter shorting out against itself, which could result in inadequate heat reaching the propellant to produce ignition. Coating on the igniter also burns when the igniter is heated, producing additional heat for rapid rocket engine ignition.

Most model rockets are reusable. But, the rocket engine (also known as rocket motor) is a non-reusable device, hence a new engine is needed for each rocket flight.

In a typical model rocket (with a strong cardboard tube as airframe), the nose cone guides the airflow around the rocket and the fins keep the rocket travelling straight. The recovery system, located inside the airframe, is used to get the rocket back safely and intact for repeated use.

After ignition, the flight starts from the launch pad (lift-off). When the powered flight ends, delay charge allows altitude gain (coasting). At the peak of the flight, the rocket arcs over (apogee) and charge pressurised inside the rocket pushes the recovery system out (ejection). Finally, the refurbishing process comes into play to fly again (recovery).

Model rocket igniter

The electrical igniter used in launching the model rocket provides enough heat to cause the temperature of the propellant to reach the desired level. When enough electric current flows through the igniter, adequate heat is available for rapid ignition of the propellant. The special insulator-igniter coating burns when it reaches a high temperature, producing extra heat for extremely rapid engine ignition.



Fig. 1: Model rocket and launchpad

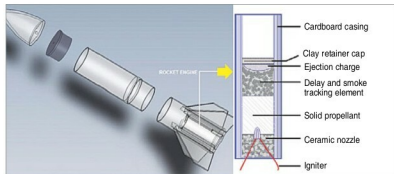


Fig. 2: Basic structure of a model rocket engine